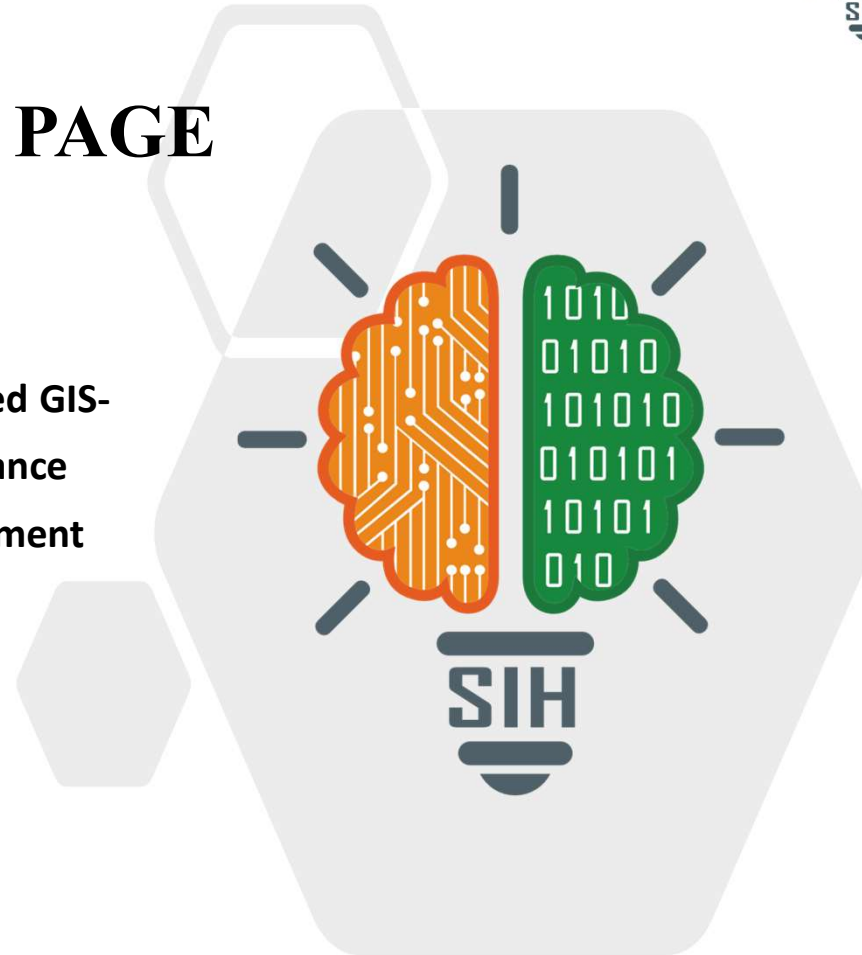


SMART INDIA HACKATHON 2026



TITLE PAGE

- **Problem Statement ID – 26014**
- **Problem Statement Title-** An Integrated GIS-based Digital Public Infrastructure for Land Governance
- **Theme-** Agriculture, FoodTech & Rural Development
- **PS Category- Software**
- **Team ID- 124203**
- **Team Name- SleepyCoderzzz**



IDEA TITLE

Our Solution

We propose a **parcel-centric Land Stack platform** that integrates fragmented land-related datasets from multiple departments through a common **GIS framework and ULPIN-based digital identity**.

How It Addresses the Problem

- **Unified Parcel View:** Connects cadastral maps, RoR, registration, zoning, building permissions, encumbrances, taxation and utilities to a common parcel.
- **Interoperability:** Uses standardized APIs and a common data model to connect diverse State and departmental systems.
- **Citizen Services:** Enables parcel search, ownership verification, transaction tracking and service requests through a single interface.
- **Data-Driven Governance:** Provides GIS dashboards and analytics for planning, monitoring and decision support.

Innovation & Uniqueness

Parcel 360° — a single view of a land parcel's ownership, rights, restrictions, transactions and services.

AI + GIS Intelligence — satellite-based change detection and AI-powered land-document extraction.

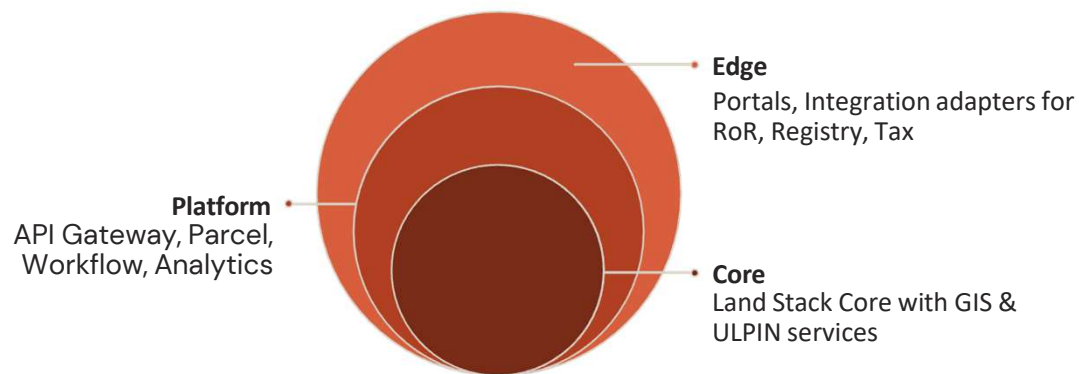
State-Adaptive Architecture — adapters map different State-specific databases and terminology into a common Land Stack model.

Core Idea: One Parcel → One Digital Identity → Multiple Connected Land Services

TECHNICAL APPROACH

Architecture, Technology & Implementation

System Architecture



The prototype demonstrates integration using **mock/sample departmental datasets** through standardized APIs, while the architecture is designed for integration with actual State and departmental systems.

Technology Stack

Frontend

Next.js, React, TypeScript, Tailwind CSS

GIS & Database

MapLibre GL JS, GeoJSON, PostgreSQL + PostGIS

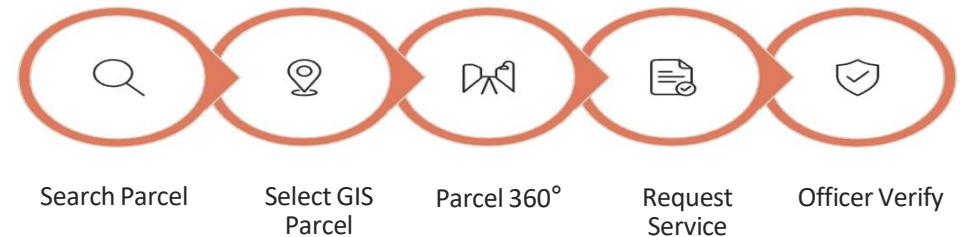
Backend & AI

Node.js, Express.js, Python, FastAPI, OCR, Computer Vision

Security & Deployment

OAuth 2.0, RBAC, JWT, TLS, Docker + Cloud Infrastructure

Prototype Flow



FEASIBILITY AND VIABILITY

Practical, Scalable & Configurable

Technical

- Mature open-source technologies and established GIS standards
- PostgreSQL/PostGIS for scalable spatial data management
- API-based architecture integrates with existing departmental systems

Operational

- Existing departmental systems continue operating unchanged
- Adapter layer handles State-specific schemas and terminology
- Role-based workflows align access with departmental responsibilities

Economic

- Open-source technologies reduce software licensing costs
- Cloud-native architecture enables incremental infrastructure scaling
- Modular deployment allows piloting in select regions first

Key Challenges & Mitigations

Challenge	Mitigation
Different State data formats	Configurable data adapters & canonical schema
Legacy departmental systems	API/adaptor-based integration
Data quality & duplication	Validation, standardization & reconciliation
Privacy & unauthorized access	RBAC, encryption & audit trails
Large geospatial datasets	PostGIS, spatial indexing & cloud storage
AI false positives	Human-in-the-loop verification

Viability Strategy: Start with one city + one village → validate the architecture → expand State-wide → replicate nationally.

IMPACT AND BENEFITS

Transforming Land Governance

Citizens

- Single-window access to land information
- Faster ownership & record verification
- Transparent transaction/service tracking
- Reduced visits to multiple departments

Government

- Interdepartmental data interoperability
- Reduced duplication and manual processes
- Complete audit trail of critical operations
- Better monitoring & administrative efficiency

Planning

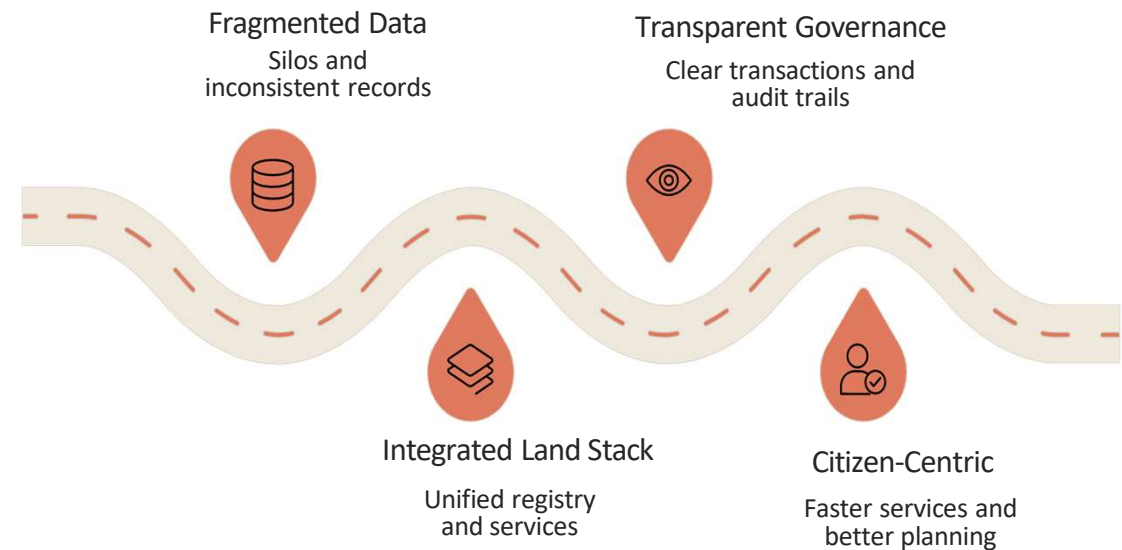
- Integrated GIS-based land intelligence
- Land-use and zoning analysis
- Infrastructure planning support
- Data-driven policy and planning

Environment

- Monitoring land-use changes via satellite
- Identification of restricted/environmental zones
- Better rural land administration
- Support for sustainable land-use planning

Economic

- Reduced administrative processing costs
- Faster land-related transactions
- Improved property-tax visibility
- Reduced information asymmetry



RESEARCH AND REFERENCES

Government / Official References

DoLR / DILRMP

Department of Land Resources, Ministry of Rural Development — Digital India Land Records Modernization Programme.

ULPIN

Department of Land Resources — Unique Land Parcel Identification Number. Reference for standardized parcel identification.

India Stack / DPI Framework

Reference for interoperable and citizen-centric digital public infrastructure.

OGC & ISO 19115

Open Geospatial Consortium standards (WMS, WFS, GeoJSON) and ISO Geographic Information Metadata for geospatial interoperability.

Technology & Research Areas

Technology References

- PostgreSQL / PostGIS Documentation
- MapLibre GL JS Documentation
- OpenStreetMap & GeoJSON Specification
- OpenAPI Specification
- OAuth 2.0 / OpenID Connect

Research Areas

- GIS-based Land Administration
- Spatial Data Infrastructure
- Digital Public Infrastructure
- Land Record Modernization
- Satellite Change Detection
- AI-based Document Intelligence
- Geospatial Interoperability

Government policies define the land-governance context; geospatial standards define interoperability; modern cloud, GIS and AI technologies provide the implementation foundation.

IMPORTANT INSTRUCTIONS



Please ensure below pointers are met while submitting the Idea PPT:

1. Kindly keep the maximum slides limit up to six **(6)**. (Including the title slide)
2. Try to avoid paragraphs and post your idea in points /diagrams / Infographics /pictures
3. Keep your explanation precise and easy to understand
4. Idea should be unique and novel.
5. You can only use provided template for making the PPT without changing the idea details pointers (mentioned in previous slides).
6. You need to save the file in PDF and upload the same on portal. No PPT, Word Doc or any other format will be supported.

Note - You can delete this slide (Important Pointers) when you upload the details of your idea on SIH portal.